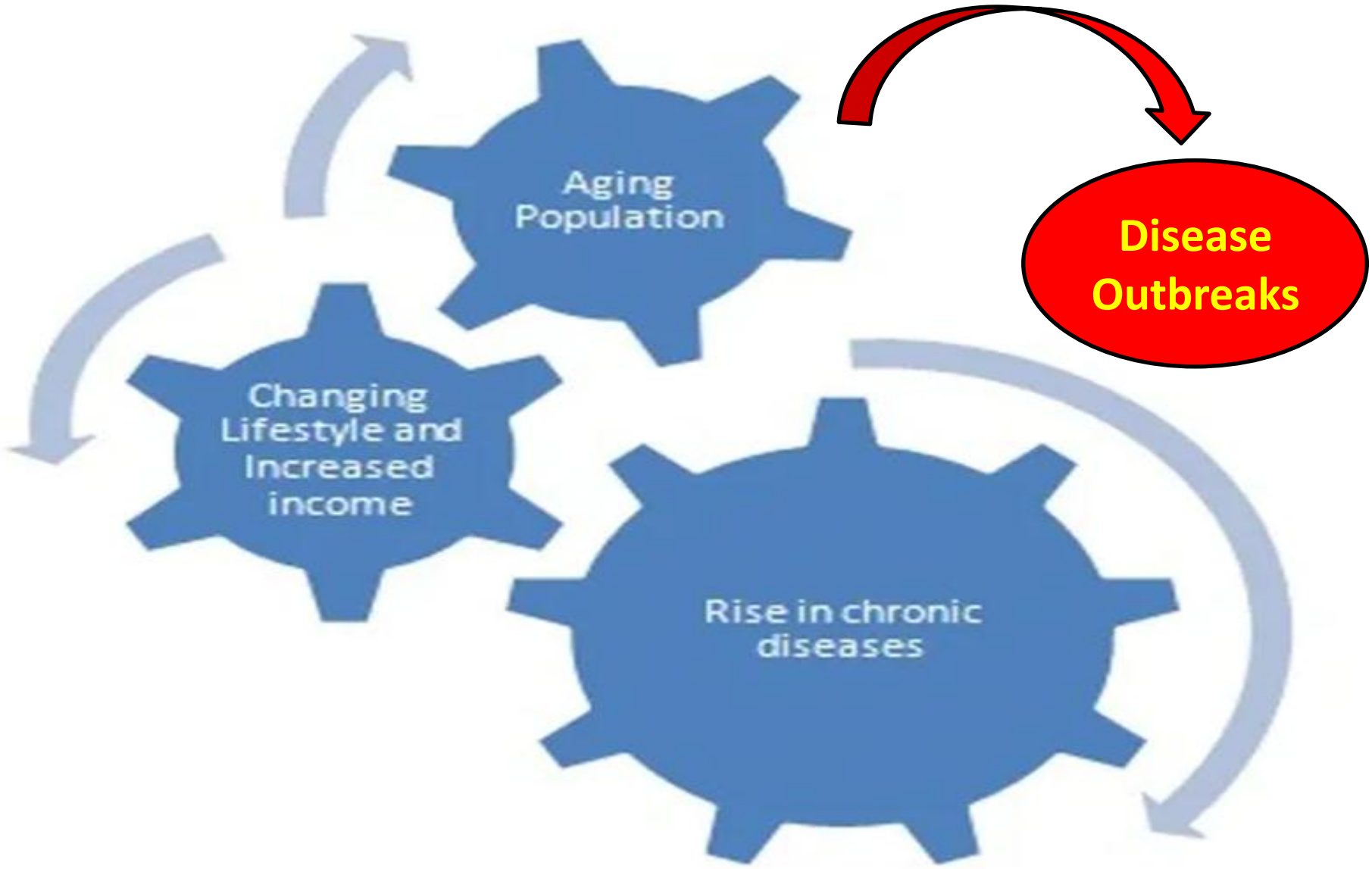


Pharmaceutical Industry and Product Life Cycle

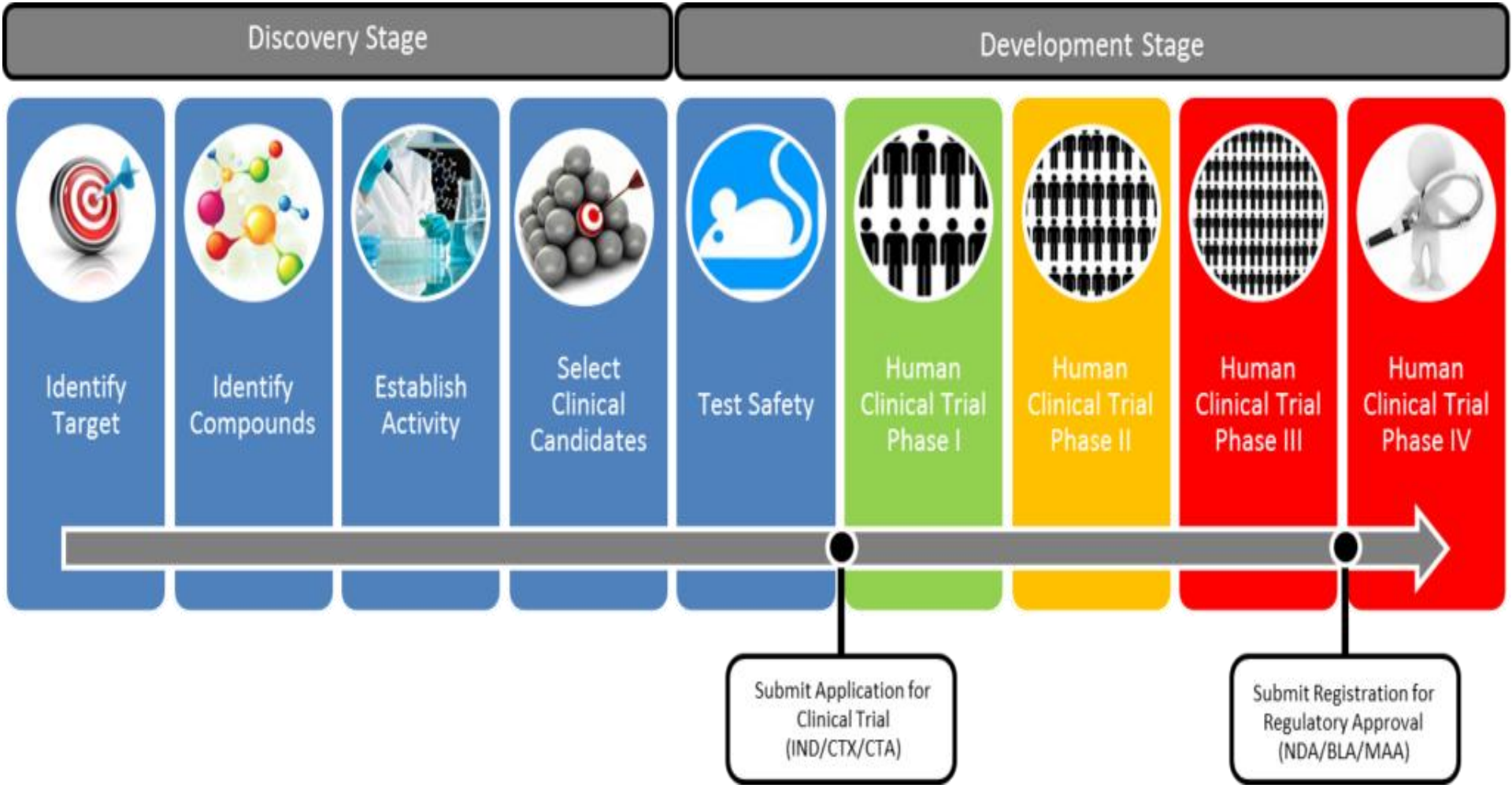
Pharmaceutical Industry

- The pharmaceutical industry discovers, develops, produces and markets drugs for human and animal use.
- The pharmaceutical industry is an essential part of the healthcare sector around the globe that deals with medications.
- The industry comprises different subfields pertaining to the development, production, and marketing of medications.
- These subfields consist of drug manufacturers, drug marketers and drug companies.

Growth Drivers for Pharma Companies



Drug Discovery and Development



DRUG DEVELOPMENT PROCESS

Out of every 10,000–15,000 new compounds identified during discovery, **five are considered safe for testing** in human volunteers. **Only one of these compounds** is typically approved **as a marketed drug.**



3-6 YEARS	6-7 YEARS	0.5-2 YEARS
Drug Discovery	Clinical Trials	FDA Review
Pre-clinical	Phase I: 20-100 Volunteers	Manufacturing
	Phase II: 100-500 Volunteers	Phase IV Post-Approval
	Phase III: 1,000-5,000 Volunteers	
Pre-discovery	IND Submitted	NDA Submitted

AVERAGE COST: $1 billion+ **DURATION:** 10-15 years*

PPD

*Source: ACRO

ppdi.com

New Product Development

- **Developing new products is a key factor for success because it shapes how the organization competes. Product development offers the opportunity to determine:**
  - **product costs**
  - **product quality**
  - **the ability to serve customers.**
- **These factors are extremely important in establishing an organization's competitive position.**
- **Product development and the design that results have a direct impact on costs because the design defines (1) how the product will be made, (2) what materials will be needed, and (3) what machines and processes will be used.**

New Product Development

- **Product quality has become a critical factor for successful competition in the international marketplace. Successfully managing quality includes:**
  - **(1) designing the product to achieve high quality**
  - **(2) designing production processes that will transform the design into a quality product**
  - **(3) developing sound training programs aimed at improving employee performance**
  - **(4) inspecting the product to determine if steps 1 through 3 have been successful.**
- **Customer service can be an important competitive advantage because meeting specific needs and providing rapid, on-time delivery are important to customers.**

New Product Development

- **This process consists of 5 phases:**
- **Phase 1: Initial Assessment**
- **An organization should continually monitor customer preferences to get ideas for new products.**
- **However, new product ideas may also come from engineering, operations, top management, or other parts of the organization.**
- **A market analysis should be made to new product ideas to forecast their probability of success.**

New Product Development

- **A market analysis for a new product might include the following steps:**
  - **Assessment of the market (current & potential customers, expected demand).**
  - **Appraisal of the competition (present competitors, their reaction to a new entry).**
  - **A preliminary market forecast (expected sales volume, expected profits).**
- **Many product ideas are discarded after such a market analysis because the expected sales are not sufficient, the estimated market price is too low to make operations profitable, or the competition is too great.**

New Product Development

- **Phase 2: Engineering and Economic Analysis:**
- **A new product idea requires careful analysis and synthesis.**
- ***Analysis* is separating a whole into its parts to determine their nature, proportion, function, and relationship.**
- ***Synthesis* is putting the parts or elements together to form a whole.**
- **Financial analysts compare the estimated profit (revenue minus cost) with the investment to determine if the product has a sufficient return (profit) for the amount invested. If this financial hurdle is crossed, a proposal for the development and testing phase.**

New Product Development

- **Phase 3:Development and Testing Phase:**
- **Engineering and operations combine to develop a product prototype. *Prototypes* are models of the product.**
- **In drug industry, a prototype takes the form of models reduced in scale.**
- **Some product ideas are discarded at this point because the prototypes cannot be made to function satisfactorily.**
- **During development and testing, a proposal for full-scale production is prepared.**

New Product Development

- **Phase 4: Final Planning:**
- **The operating and marketing plans and the financial analysis are presented to top management for approval.**
- **Many projects never reach the next phase because of poor profit potential.**

New Product Development

- **Phase 5: Launching the Product and Market Surveys:**
- **Management approval begins the implementation of the marketing and operating plans. Facilities are constructed, equipment is purchased, people are hired and trained, and materials are acquired.**

Drug Regulation and Approval

- **There a number of regulatory bodies internationally and locally that regulate pharmaceutical industry and monitor drug patenting, R&D, safety, quality, efficacy, testing, pricing and marketing of drugs.**
  - **World Health Organization (WHO)**
  - **US Food and Drug Administration (FDA)**
  - **Medicines and Healthcare Products Regulatory Agency (MHRA)**
  - **European Medicines Agency (EMA)**
  - **Egypt: Egyptian Drug Agency (EDA)**

Drugs/Medicines

- **Drugs have a great positive influence on global health and economic productivity by:**
  - **saving lives**
  - **increasing life spans**
  - **reducing suffering**
  - **preventing surgeries**
  - **shortening hospital stays**
- **The healthcare sector consists of businesses that provide drugs, medical services and medical device/equipment.**

Differences between Products and Services

Physical goods	Services
tangible	intangible
homogeneous	heterogeneous
Production and distribution are separated from consumption	Production, distribution and consumption are simultaneous processes
A thing	An activity or process
Core value processed in factory	Core value produced in the buyer-seller interaction
Customers do not participate in the production process	Customers participate in production
Can be kept in stock	Cannot be kept in stock
Transfer of ownership	No transfer of ownership

Examples

- **Prescription drugs**
- **OTC drugs**
- **Dietary supplements**
- **Cosmetics/medical device**

- **Patient counseling**
- **Health promotion**
- **Drug information**
- **Medication management**

Pharmaceutical Market	FMCG market Fast-Moving Consumer Goods products
	
Supplies drugs or medicinal products used for diagnosis, prevention or treatment of specific medical conditions.	Supplies consumer products that are routinely used (ie toothpaste, snack foods, stationary)
REGULATIONS	
Highly Regulated Manufacturing regulations Sold by registered pharmacist Prescribed by a registered medical practitioner	UNREGULATED Regulations do not apply in the manufacturing, supply or use of FMCG products
AVAILBILITY	
Can only be sold by registered pharmacists and prescribed by registered medical practitioners *Prescription drugs* The doctor is the customer and the patient is the consumer *OTC drugs* The pharmacist may be the customer and the patient is the consumer	Can be sold in any general store and be purchased by any consumer without any interference of any authority or law There is no difference between the customer and the consumer
PRICE	
Regulated by authorities (TGA)	There is no regulation on the price of FMCG
ADVERTSING	
Advertisements of pharmaceutical products are regulated by authorities. Prescription drugs cannot be advertised on television, radio, social media or in public places.	Although basic guidelines need to be followed, there are no regulations in relation to the advertisement of FMCG.



Branded vs. Generic Drugs

Branded vs. Generic Drugs

- **Pharmaceutical sales are comprised of two types of drugs: brand-named (originator, innovator) and generic drugs.**

- **Brand-name drugs/innovator/originator rely on patents, regulatory data protection and other forms of intellectual property rights (IPR) to justify the investment required to bring a product to market.**

- **Generic drugs are copies of brand-name drugs that must have the same active ingredients , quality, strength, safety, dosage form and route of administration.**

Branded Medicines and Generic Medicines

- **Similarities:**
  - It must contain the same active ingredient
  - It must have the same dosage strength
  - It must have the same dosage form

- **Differences:**
  - They have different names
  - They have different sizes, shapes, colors or markings
  - Generic costs less than brand-name drugs

BRAND NAME DRUG
NDA REQUIREMENT

New Drug Application

1. Labeling
2. Pharma
3. Chemistry
4. Manufacturing
5. Controls
6. Microbiology
7. Testing
8. Animal studies
9. Clinical studies
10. Bioavailability

Supplied by one company
Sold under trademarked name

GENERIC DRUG
ANDA REQUIREMENT

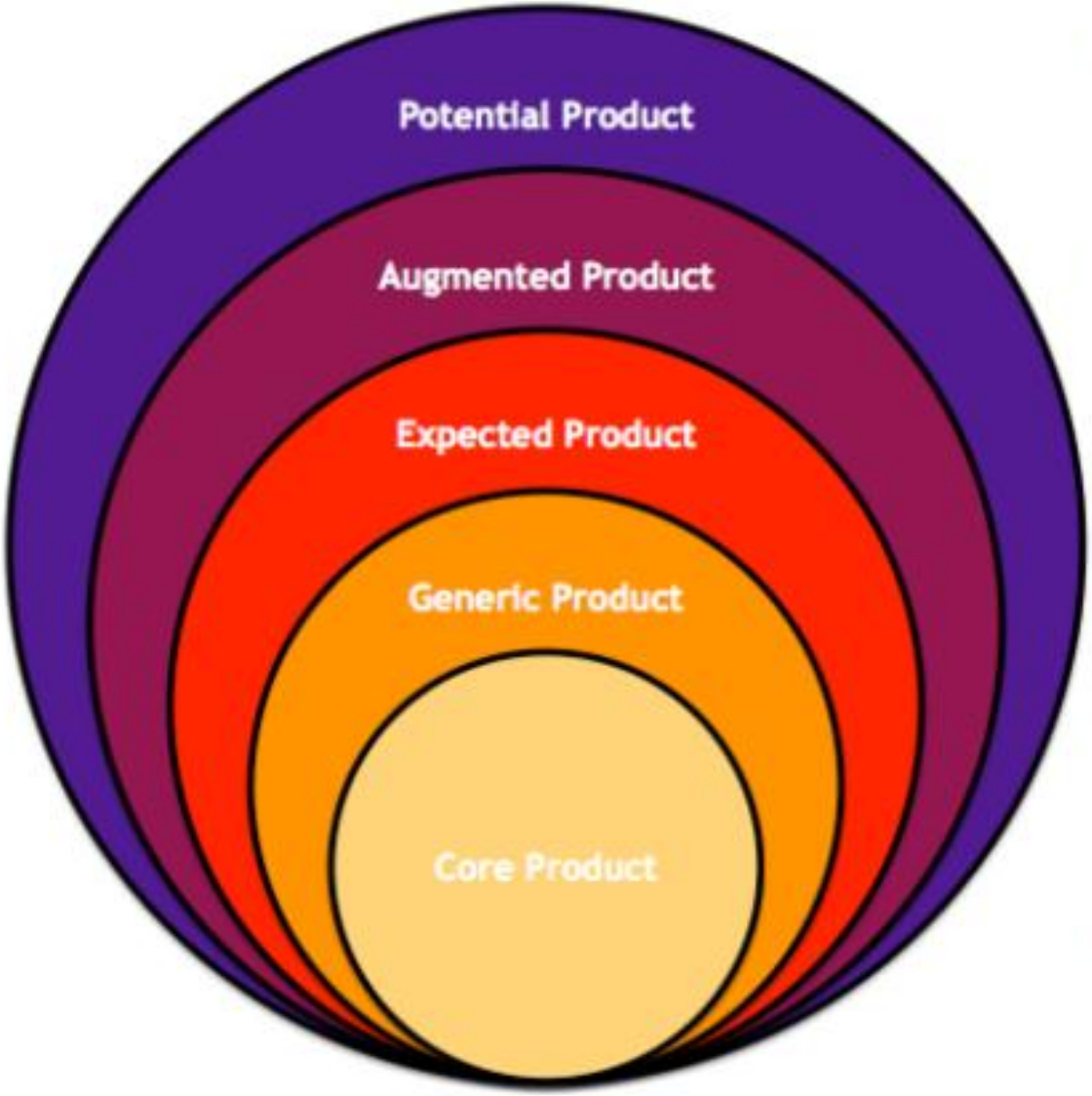
Abbreviated New Drug Application

1. Labeling
2. Pharma
3. Chemistry
4. Manufacturing
5. Control
6. Microbiology
7. Testing
8. Bioequivalence



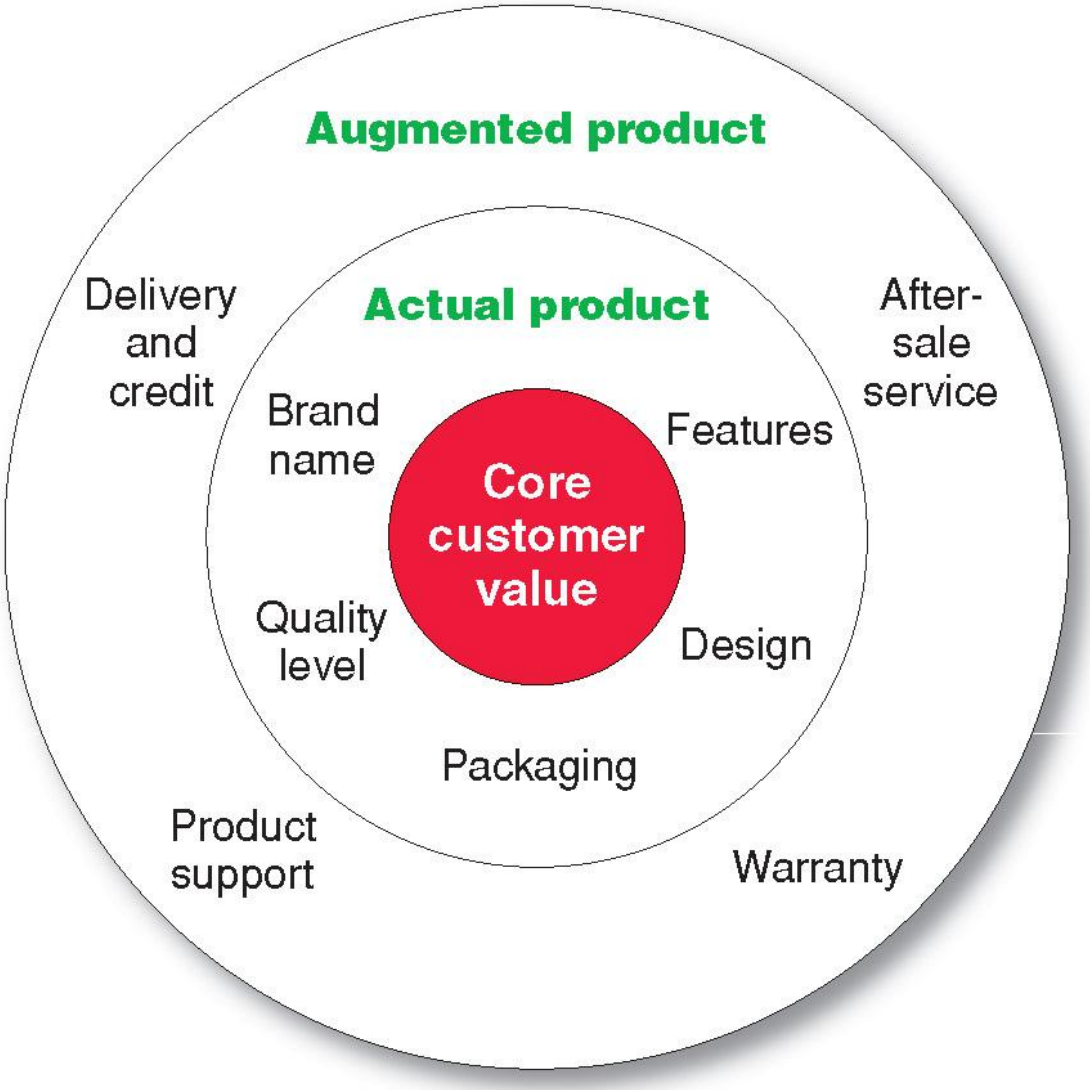
Supplied by more than company
Sold under active ingredient name

Product Levels According To Philip Kotler



- **Potential product** provides additional tangible and intangible features.
- **Augmented product** gives more than physical product and sets it apart from competitors.
- **Expected product** offers generic product plus other attributes consumers want.
- **Generic product** provides actual product with tangible qualities.
- **Core product** fulfills basic benefit consumers want.

The Concept of Product: Product Anatomy
Levels of Product (Three Levels)



PRODUCT ATTRIBUTES

Product attributes can be explain in three aspects:

1) Product Quality
2) Product features
3) Product Style & Design





Product Characteristics

- **A full understanding of the product's characteristic:**
- **Features: the specific mechanism of action, dosage form, shape or color.**
- **Attributes: things that reside in the product including how it functions -the speed of onset of action, side effects, such as drowsiness, single dose rather than multiple doses.**
- **Benefits: expressed in terms of what the doctor or patient gets from the product.**
- **Limitations: must be clearly indicated in all materials and discussions with the doctor. The potential risk of adverse reactions of a medicine, contraindications and drug interactions**

Colors for Pharmaceutical Products

- **Colors for pharmaceutical products are used to impart organoleptic properties or technical purposes.**
- **Organoleptic properties are aspects of food or other substances that make it more appealing to the senses of consumers such as sight, smell, taste and touch.**
- **Colors are combined with flavors such as strawberry, mint etc. and matched with the formulation of the drugs to be created.**

Design Attractive Product Packaging

- **Packaging is the first step to getting your new product noticed.**
- **Create packaging that is bold, right color scheme, pleasing to the eye and memorable.**
- **Consider including cusp/tip shapes in your packaging.**
- **Cusp/Tip shapes are pointy and sharp on packaging, so they make the customers people to take a second look.**
- **Stick to something simple.**
- **Busy designs may confuse customers, so having a simple looking package may also be a good way to go.**

Key Strategies for New Product Development

- **Five parameters are deciding factors in development of a new product:**
  - **product best fit for customer**
  - **innovative product**
  - **low service cost**
  - **low product cost**
  - **product first to market**
- **Study of Product Life Cycle reveals that customers' viewpoints are the key factors for manufacturing a product fit for customer requirements.**

Create the Perfect Product Slogan (Catchphrase)

- **Choose a phrase that is catchy, captures the essence of your product and will speak to your target customer.**
- **The slogan should consist of simple language and could contain words beginning with the same letter to make it catchier.**
- **Popular slogans include McDonald's "I'm lovin' it"; Nike's "Just do it" and Walmart's "Save Money. Live Better."**
- **One study found that the three main factors in creating a great slogan are clarity, creativity and familiarity.**
- **Consider hiring a writer or communications specialist or someone with a remarkable command of language to come up with this slogan.**

Summary of Product Characteristics

- The summary of product characteristics contains a description of a certain medicinal product's properties and the conditions attached to its use.
- Example:
  - Name
  - Composition (declaration)
  - Pharmaceutical form and strength
  - Authorised applications (indications)
  - Adverse reactions
  - Cautions and safety regulations
  - Shelf-life
  - Storage conditions
  - Holder of marketing authorisation (firm)

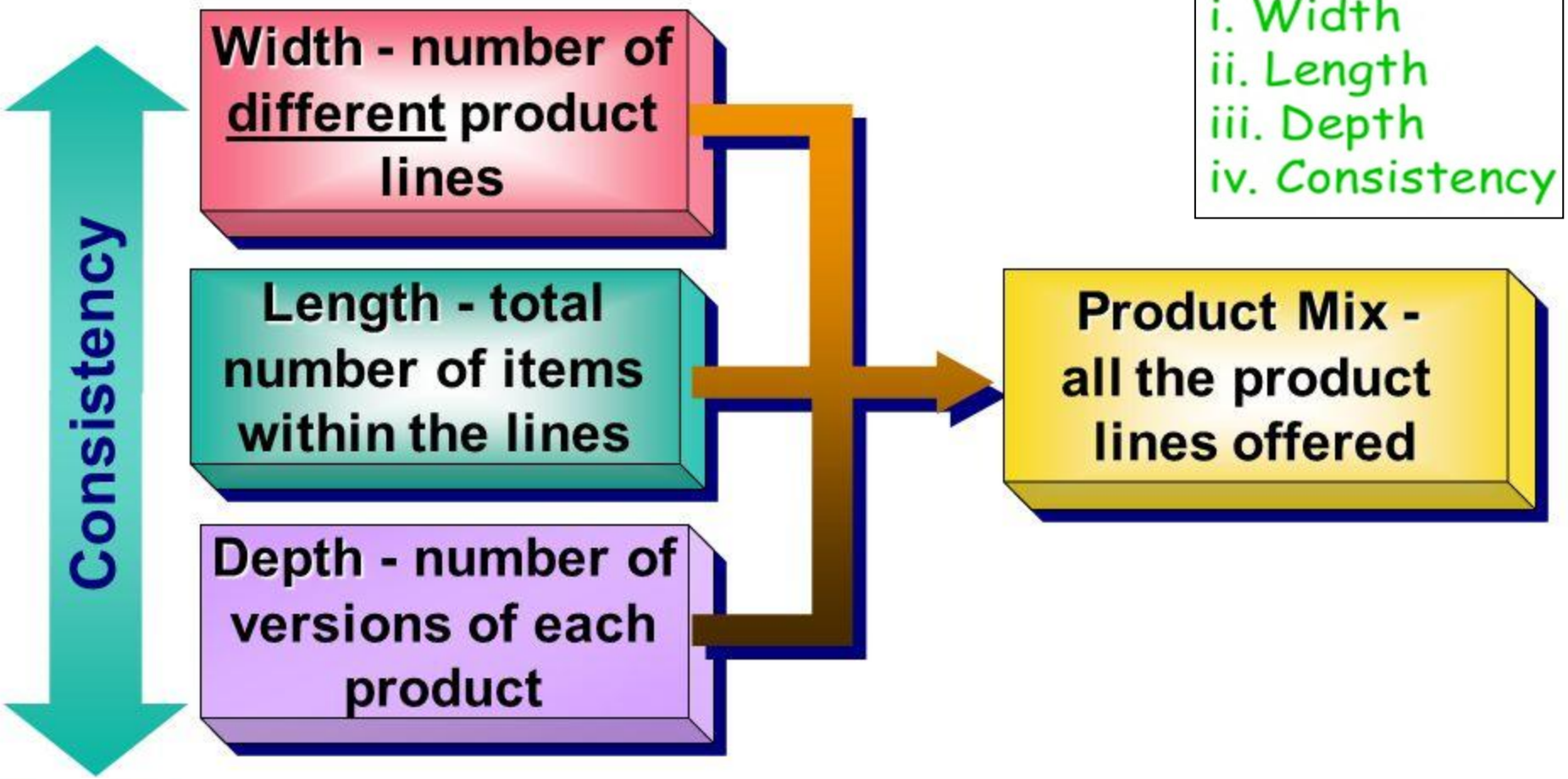


Product Mix

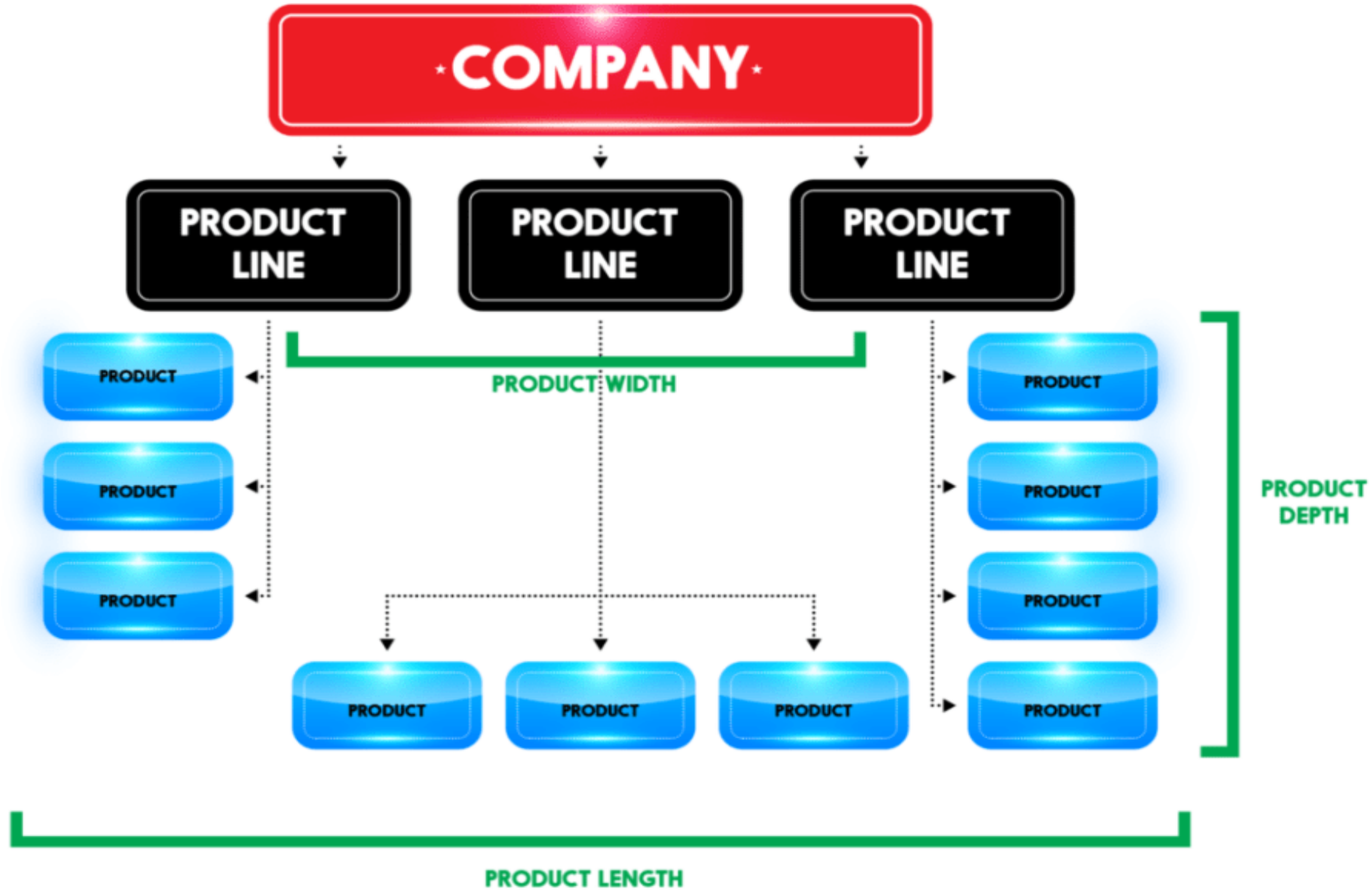
Product Mix: Products of a company

Any company's product mix has 4 dimensions:

i. Width
ii. Length
iii. Depth
iv. Consistency



Consistency : Closely related the various product lines.



PRODUCT MIX WILL BE CONSISTENT IF ALL THE PRODUCTS IN ALL THE PRODUCT LINES ARE MORE OR LESS SIMILAR


Launch a New
Pharmaceutical Product
WHERE
WHY
WHEN
?
WHO
HOW
WHAT

Launch a New Pharmaceutical Product

- **Dundas and Krentler (1982): launch strategies are tools that are used to guide new product launches and determine the performance of a new product.**
- **Hultink and Hart: Launching strategies describes those marketing decisions that are necessary to position a new product to its target market.**

- **The new product launch phase is a critical part of the total new product development process.**
- **The quality of launch programs greatly impacts the success of product introduction.**

Launch a New Pharmaceutical Product

- **The pharmaceutical industry finds itself at crossroads where dozens of prescription drugs will lose patent protection within the next few years and new drug launches have been hit-or-miss.**
- **It is a challenge for companies to create prescription drugs that can meet higher payer standards and that can differentiate themselves from the crowded marketplace based on areas such as safety, efficacy and pricing.**
- **While prescription drug launches always held an air of unpredictability, companies now seem to struggle even more with launch execution in today's complex world when failure is more expensive than developing a new drug costs.**

Launch a New Pharmaceutical Product

- **Launching a new product informs the public about your product.**
- **New product development is a very risky business that the product launch itself requires the largest commitment of resources.**

Brand Extension

- Many companies adopt brand extension as strategy with the aim of benefiting from the brand knowledge achieved in the current market.
- Brand extension as a marketing strategy has become even more attractive in today's environment where developing a new product costs a lot of money and can be time consuming.

Orchestrate a Rolling/Orderly Launch

- A rolling launch should begin six to eight weeks before the final product opening.
- A plenty of “leaks” regarding your product as the opening date draws closer.
- "Leaked” photos or tweets regarding your product create an atmosphere of story that builds interest.
- Introducing/telling “news” stories about consumers are responding or trials can create motivation.

Causes of Unsuccessful Product Launch



- Poor launch time
- Poor launching methods
- Product attributes fail to address patient needs
- Price of product
- Insufficient sale force
- Inadequate budget
- Ineffective marketing strategies





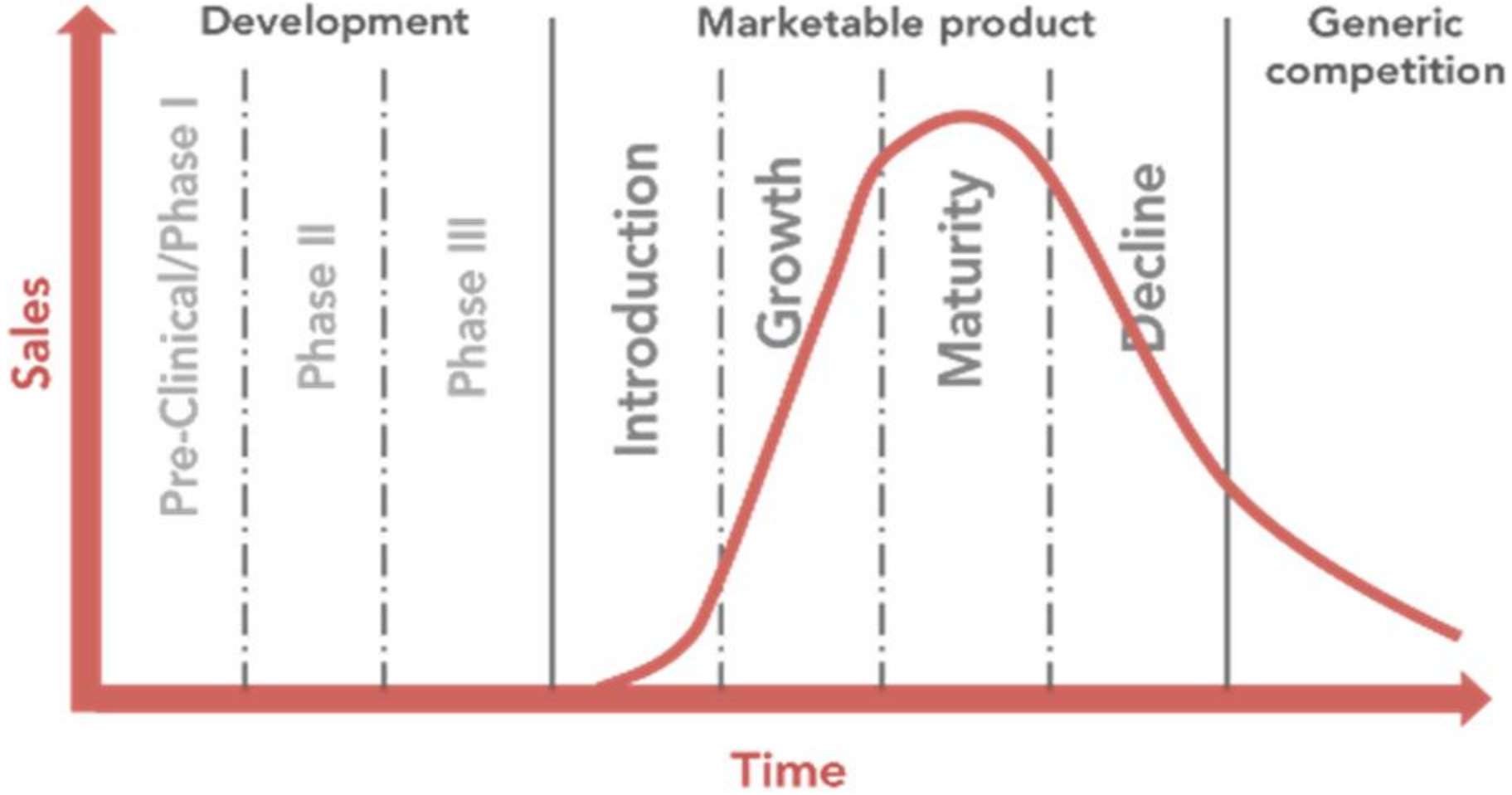




Product Life Cycle (PLC)

- All products and services have certain life cycles.
- Product life cycle (PLC) refers to the entire life cycle of a pharmaceutical product from its conception, through design and manufacture, launch to service and withdrawal.
- The understanding of a product's life cycle can help a company to understand its position in the market compared to competitors and the product's success or failure.
- The typical PLC consists of five main aspects:
  - Development
  - Introduction
  - Growth
  - Maturity
  - Decline

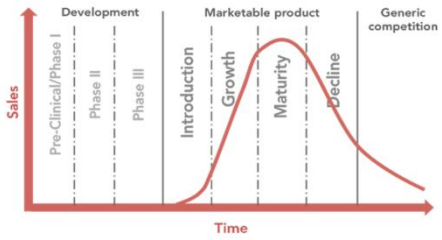
Product Life Cycle (PLC)
Product Management Cycle (PLM)



Product Life Cycle

- **Product life cycle is the status of a product in the company portfolio.**
- **Pharmaceutical products tend to go through different stages, each stage being affected by different competitive conditions.**
- **These stages require different marketing strategies at different times if sales and profits are of concern.**
- **The length of a product's life cycle is not a fixed period of time.**
- **It can last from weeks to years depending on the type of product.**
- **A market may be defined as a place where buyers and sellers meet, products or services are offered for sale and transfers of ownership occur.**

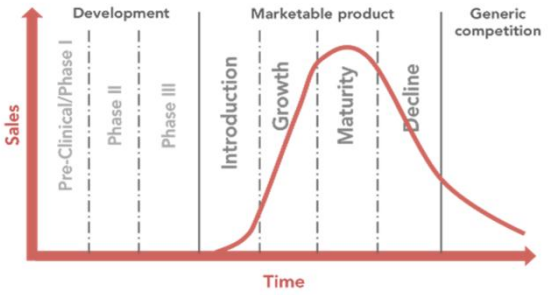
Introduction Stage



- Product launched into the market.
- Sales grow slowly.
- Informative advertising is done.
- Firm might not earn a profit at this stage.
- Price skimming may be used if the product is new invention and has no competitors.
- Competitive pricing may be used if it already has lot of competitors.

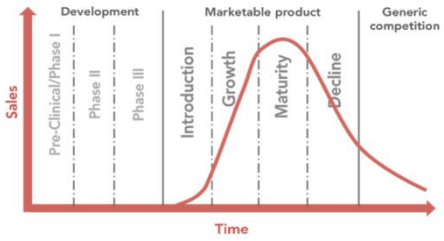
Growth Stage

- Sales grow rapidly.
- Persuasive advertising may be used to increase sales.
- Prices may be reduced if faced by stiff competition.
- Company starts earning profits.

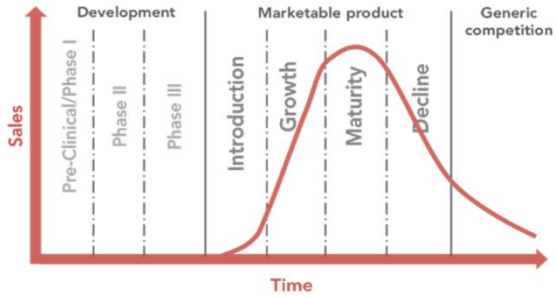


Maturity/Saturation Stage

- Sales are stagnant/stationary.
- Profits are at the highest level.
- Maximum competition but no new competitors and the market is already crowded with the same types of products.
- Advertising efforts at its highest point.
- Promotional pricing or competitive pricing may be a good choice.



Decline Stage



- Sales start to decline.
- Profits start to fall.
- Advertising is reduced.
- Marketing research to find out whether this decline is permanent or temporary.
- If the decline is permanent then stop the production of the product, otherwise implement extension strategies.

Extension Stage

- Introduce new variations of the original product.
- Make small changes in the color, design or packaging.
- Start a new advertising campaign.
- Add more selling outlets to boost sales.
- Try to sell the product in different markets.

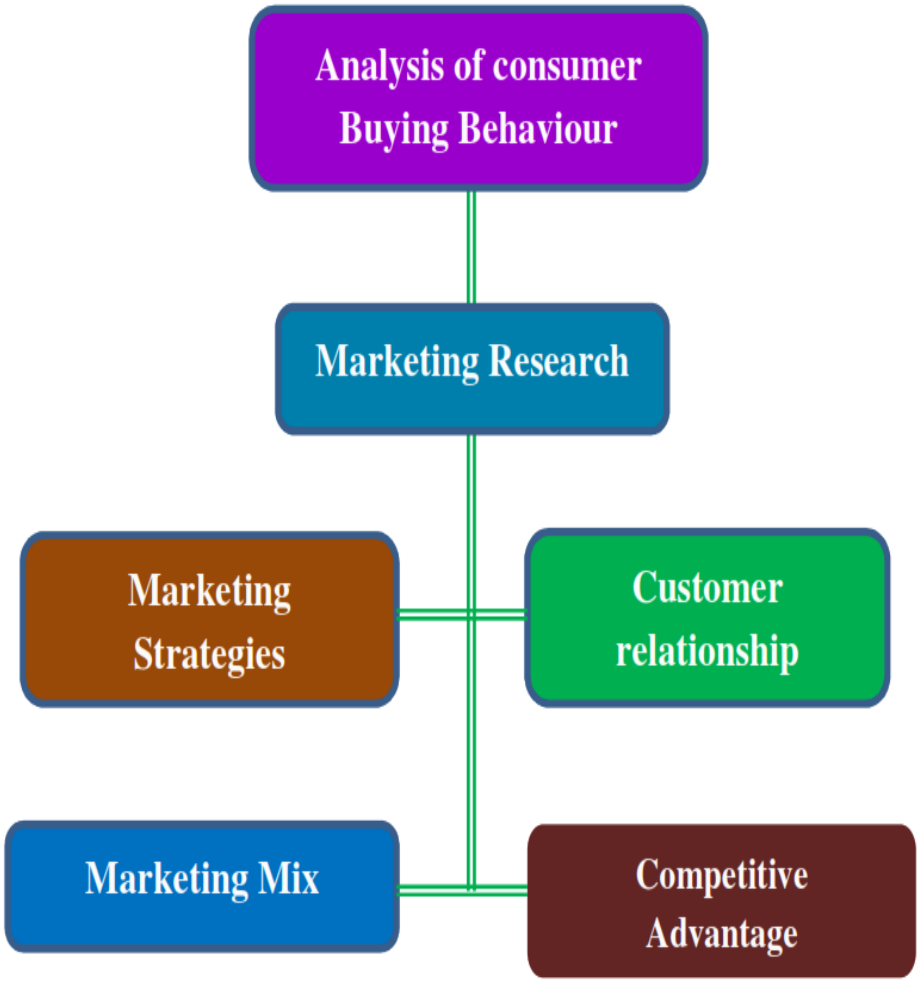
Consumer Buyer Behavior

- **Consumer/customer is the 'King' of the market and every marketer aims at identifying the buying behaviour and to increase the satisfaction level by fulfilling the needs and wants of consumer.**
- **Consumer buying behaviour concept explains about how the individuals, groups or organizations do selection, buying, using or disposing of the ideas, goods and services, their experiences for the satisfaction of need and want.**

Consumer Buyer Behavior

- **Consumer can be any natural person, group or an organization who are involved in the process of buying the goods and services for the personal use.**
- **Consumers are mainly influenced by the marketing strategies applied by the organization in the form of advertising, promotional tools, etc**

Consumer Buyer Behavior

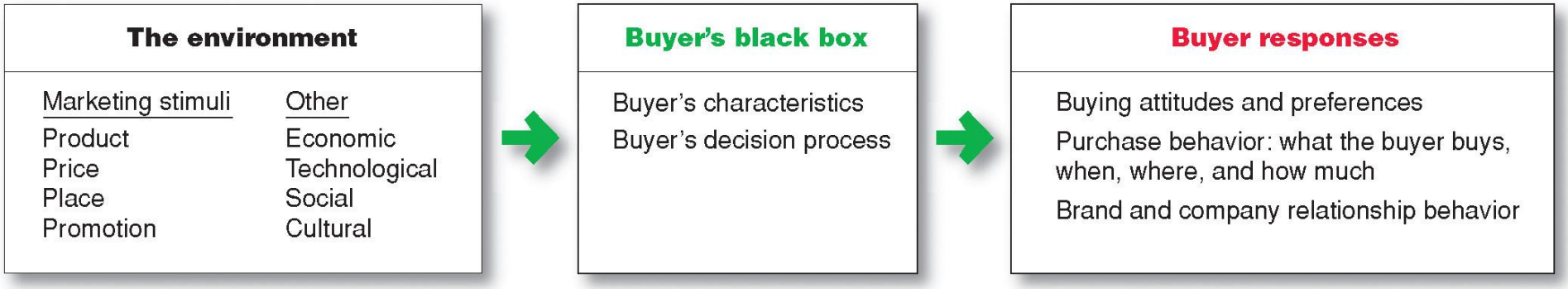


Theoretical framework of consumer buying behavior

Consumer Buying Behaviour

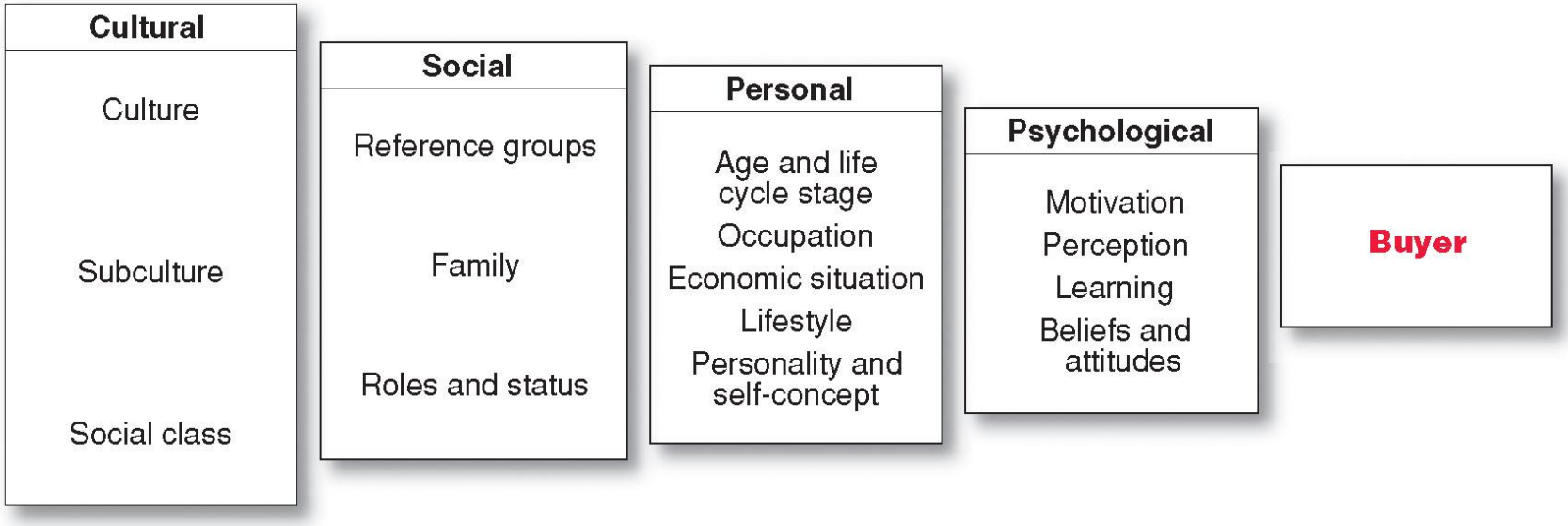
- **Consumer buyer behavior: the buying behavior of final consumers, individuals and households, who buy goods and services for personal consumption**
- **Buying behavior of the customer is nothing but decision making process or the acts that are involved in buying and using of the purchased goods".**
- **The more the number of buyers of the product, the more is the success of the firm who produce them.**
- **Company should undertake a survey to understand the needs of the customer and should manufacture the goods which best satisfies such need.**

Model of Consumer Behavior



Characteristics Affecting Consumer Behavior

Factors Influencing Consumer Behavior



Characteristics Affecting Consumer Behavior

Cultural Factors

- Culture is the values, perceptions, wants, attitudes and behavior from family and other societies.
- It also regulates what should be advertised and sold in the market.
- It determines the way people live in society with regard to eating habits, wearing clothes, residing, travelling, watching and entertainment.
- These factors are affecting the social life and international market on a big role.

Characteristics Affecting Consumer Behavior

Social Factors

- Social classes are society permanent and ordered divisions whose members share similar values, interests, and behaviors
- Measured by a combination of occupation, income, education, wealth, and other variables.
- Family is the most important consumer-buying organization in society
- Social roles and status are the groups, family, clubs, and organizations that a person belongs to that can define role and social status

Characteristics Affecting Consumer Behavior

Personal Factors

- Personality refers to the unique psychological characteristics that lead to consistent and lasting responses to the consumer's environment
- Age and life-cycle stage
  - Youth
  - Adults
  - Old
- Occupation affects the goods and services bought by consumers

Characteristics Affecting Consumer Behavior

Psychological Factors

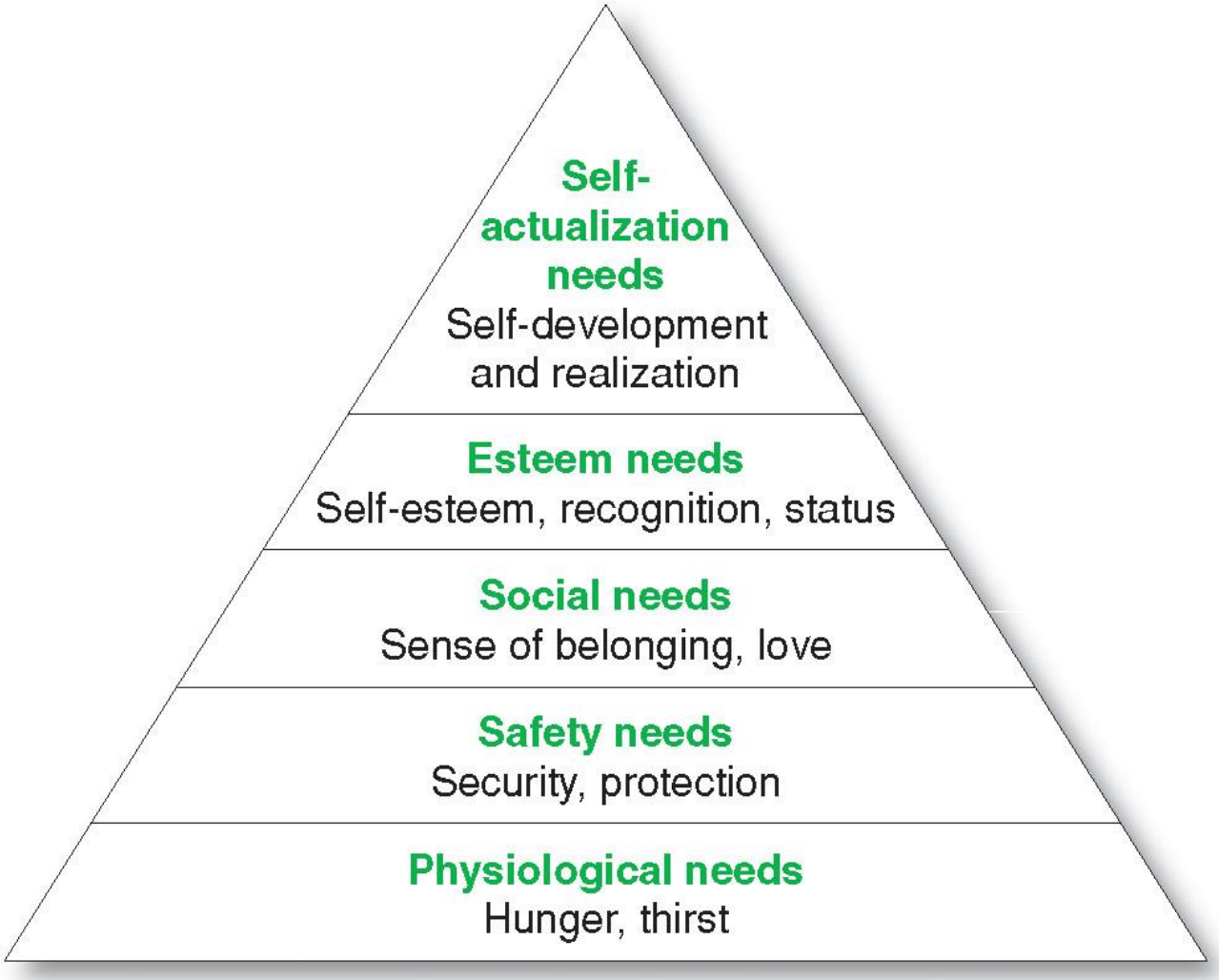
- Motivation
- Perception
- Learning
- Beliefs and attitudes

Characteristics Affecting Consumer Behavior

Psychological Factors
Motivation

- A motive is a need that is sufficiently pressing to direct the person to seek satisfaction
- Motivation research refers to qualitative research designed to probe consumers' hidden, subconscious motivations

Characteristics Affecting Consumer Behavior



Maslow's Hierarchy of Needs

Characteristics Affecting Consumer Behavior

Psychological Factors

- **Learning is the change in an individual's behavior arising from experience and occurs through interplay of drives, stimuli, cues and responses.**

Characteristics Affecting Consumer Behavior

Psychological Factors
Beliefs and Attitudes

- **Belief is a descriptive thought that a person has about something based on:**
  - **Knowledge**
  - **Opinion**
  - **Faith**
- **Attitudes describe a person's relatively consistent evaluations, feelings, and tendencies toward an object or idea.**

Types of Consumer Buying Behaviour

- Consumers decision making changes based on the type of the product purchase which is accordingly as per the need.
- The marketer should be aware of the same for the organization profits and growth.
- The decisions with regard to buying soap, bat, mobile phone or car is different and it changes.
- Usually, the complicated and highly expensive product need more deliberate involvement of the buyer.
- As shown in the figure no. 2.4 of buying behaviour types of consumer is depended on the degree of buyer involvement and differences of degree among the various brands

Types of Buying Decision Behavior

- Complex buying behavior
- Dissonance-reducing buying behavior
- Habitual buying behavior
- Variety-seeking buying behavior

Types of Buying Decision Behavior

Four Types of Buying Behavior

	High involvement	Low involvement
Significant differences between brands	Complex buying behavior	Variety-seeking buying behavior
Few differences between brands	Dissonance-reducing buying behavior	Habitual buying behavior

Complex Buying Behaviour

- This type of buying behaviour includes high involvement buying products like marble or granite for wall cladding or kitchen top.
- To purchase such kind of products buyer may not be aware of what attributes to be considered due to which he/she may does the research.
- In such a situation marketers can educate or make aware the consumer regarding the product attributes, should motivate sales personnel to influence the consumer in buying the product as per the brand choice.

Variety Seeking Behaviour

- **The type itself indicates the products where the consumers prefer variety while buying the product like biscuits.**
- **Consumers often switch brands of such category because it requires low – involvement during the purchase decision making.**
- **Marketer should develop habitual buying by keeping the product always in stock or marinating of stock near to payment counters or making of frequent reminders for the purchase.**
- **For such products marketer will influence the buyer by using promotional tools like discount, coupons, free samples, variety offers, etc.**

Dissonance – Reducing Buying Behaviour

- This type of buying behaviour is applicable to the high – involvement goods like purchase of marble or granite tiles or slabs which are quiet costly and self-expressive.
- In such a case the buyer may compare the similar products with the competitor one as per the price range of the product.
- After purchasing the product sometimes the consumer may feel dissonance or disagreement for the brand after hearing the advantages of the competitor brand or disadvantages of the product purchased.
- In this situation the marketer should supply the belief and evaluation to make the consumer feel good about the product or brand chosen by him or her.

Habitual Buying Behaviour

- This type of buying behaviour is applicable to the low – involvement products like salt or floor cleaner.
- In such type of brands the consumer preference for the buying of the goods is due to habits of choice and not due to strong brand loyalty towards that brand.
- They act as unreceptive listeners for the information provided by the enterprises through advertisements.
- Continuous advertisement increases the brand familiarity among the brands of frequent use rather the brand - Conviction.
- To motivate such customers' marketers should try to use sales promotion tools like 'Buy one Get One Free' to attract for the purchase.

Buyer Decision Process

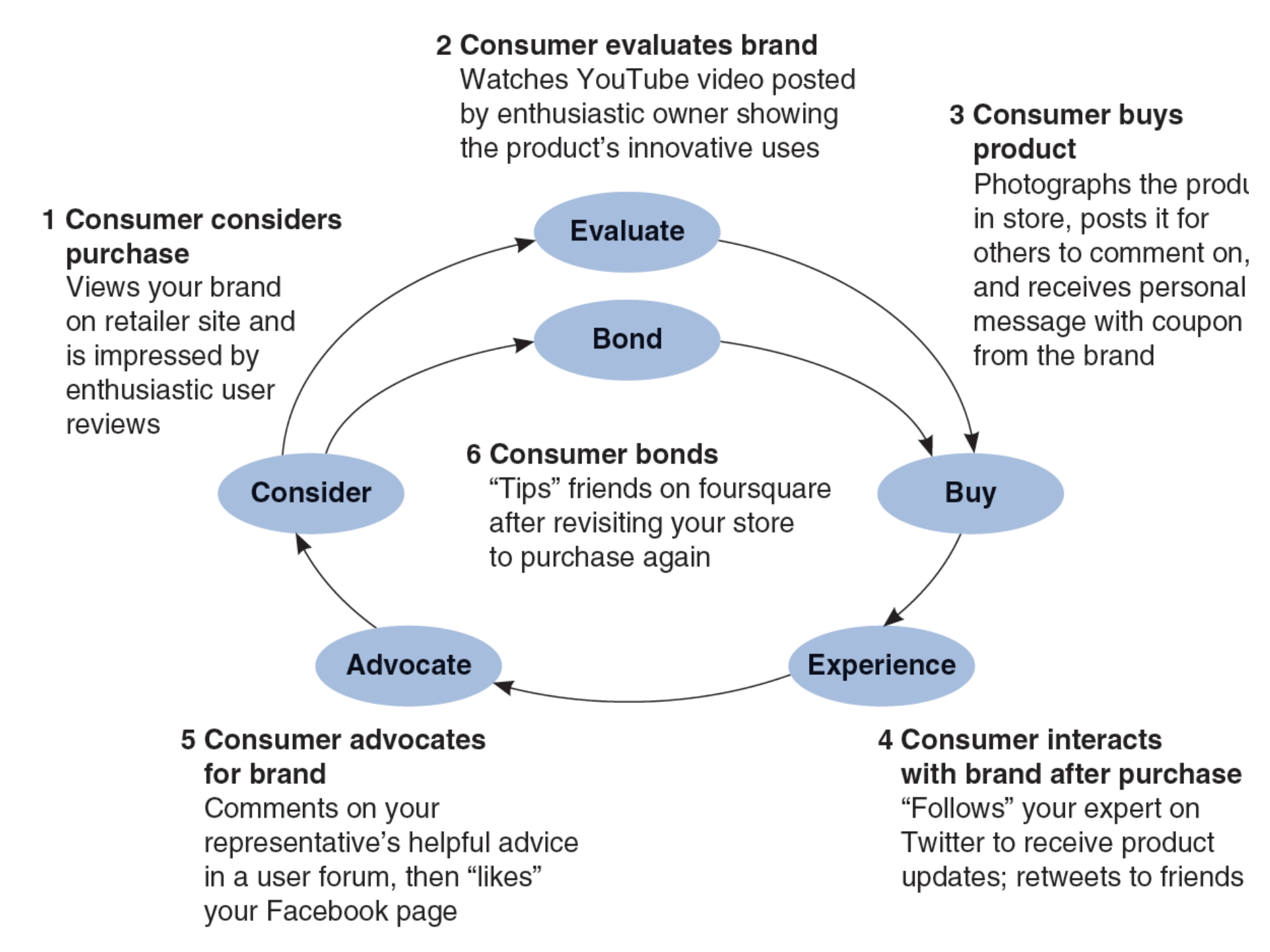
- Over the past decade, the environment in which consumers evaluate brands and make purchase decisions has changes as social media has become pervasive in our daily lives and is influencing consumer behaviour
- Consumers can access and retrieve information, connect with one another to share it, discuss products/services, and interact with marketers
- Consumers now don’t make purchase decisions in the linear manner depicted by the traditional models, but go through iterative process

Buyer Decision Process

Buyer Decision Making Process

- **'Buyer Decision Process' can be referred to as the decision making process carried on in respect of the potential market transaction before, during and after the buying of the goods or services'.**




2 Consumer evaluates brand
Watches YouTube video posted by enthusiastic owner showing the product's innovative uses
3 Consumer buys product
Photographs the produ in store, posts it for others to comment on, and receives personal message with coupon from the brand
1 Consumer considers purchase
Views your brand on retailer site and is impressed by enthusiastic user reviews
Evaluate
Bond
Consider
Buy
6 Consumer bonds
"Tips" friends on foursquare after revisiting your store to purchase again
Advocate
Experience
5 Consumer advocates for brand
Comments on your representative's helpful advice in a user forum, then "likes" your Facebook page
4 Consumer interacts with brand after purchase
"Follows" your expert on Twitter to receive product updates; retweets to friends

Buyer Decision Process

Need Recognition

- **Occurs when the buyer recognizes a problem or need triggered by:**
  - **Internal stimuli**
  - **External stimuli**

Buyer Decision Process

Information Search
Sources of Information

- Personal sources—family and friends
- Commercial sources—advertising, Internet
- Public sources—mass media, consumer organizations
- Experiential sources—handling, examining, using the product

Buyer Decision Process

Evaluation of Alternatives

- **How the consumer processes information to arrive at brand choices**

Buyer Decision Process

Purchase Decision

- The act by the consumer to buy the most preferred brand
- The purchase decision can be affected by:
  - Attitudes of others
  - Unexpected situational factors

Buyer Decision Process

Post-Purchase Decision

- Customer satisfaction is a key to building profitable relationships with consumers—to keeping and growing consumers and reaping their customer lifetime value
- The satisfaction or dissatisfaction that the consumer feels about the purchase
- Relationship between:
  - Consumer's expectations
  - Product's performance
- The larger the gap between expectation and performance, the greater the consumer's dissatisfaction
- Cognitive dissonance is the discomfort caused by a post-purchase conflict

Buyer Decision Process for New Products

- Adoption process is the mental process an individual goes through from first learning about an innovation to final regular use.
- Stages in the process include:



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